1. A regular, periodic three-dimensional mathematical array of points in space where every point has identical surroundings is a:

(A) Space Lattice
(B) Unit Cell
(C) Primitive Cell
(D) Amorphous Matrix

2. The fundamental physical relationship defining a crystal structure is mathematically expressed as:

(A) Crystal Structure = Lattice + Basis
(B) Crystal Structure = Unit Cell * Volume
(C) Crystal Structure = Lattice / Basis
(D) Crystal Structure = Primitive Cell + Grain

3. The atom, ion, or collection of atoms attached identically to each lattice point is known as the:

(A) Basis (or Motif)
(B) Unit Vector
(C) Miller Index
(D) Lattice Constant

4. The smallest repeating geometric volumetric unit that generates the entire crystal lattice through translation is the:

(A) Unit Cell
(B) Primitive Cell
(C) Wigner-Seitz Cell
(D) Brillouin Zone

5. A unit cell that contains strictly ONE constituent lattice point is specifically classified as a:

(A) Primitive Cell
(B) Non-primitive Cell
(C) Conventional Cell
(D) Supercell

6. In three-dimensional space, the total number of distinct Bravais lattices is exactly:

(A) 14
(B) 7
(C) 32
(D) 230

7. All three-dimensional crystals can be categorized into how many fundamental Crystal Systems?

(A) 7
(B) 14
(C) 6
(D) 12

8. For a cubic crystal system, the axial lengths and interaxial angles satisfy:

(A) a = b = c, and alpha = beta = gamma = 90 degrees
(B) a = b != c, and alpha = beta = gamma = 90 degrees
(C) a != b != c, and alpha = beta = gamma = 90 degrees
(D) a = b = c, and alpha = beta = gamma != 90 degrees

9. For a tetragonal crystal system, the lattice parameters satisfy:

(A) a = b != c, and alpha = beta = gamma = 90 degrees
(B) a = b = c, and alpha = beta = gamma = 90 degrees
(C) a != b != c, and alpha = beta = gamma = 90 degrees
(D) a = b = c, and alpha != beta != gamma

10. For an orthorhombic crystal system, the lattice parameters satisfy:

(A) a != b != c, and alpha = beta = gamma = 90 degrees
(B) a = b = c, and alpha = beta = gamma = 90 degrees
(C) a = b != c, and alpha = beta = gamma = 90 degrees
(D) a != b != c, and alpha != beta != gamma

11. The crystal system exhibiting the lowest degree of symmetry (a != b != c and alpha != beta != gamma != 90 deg) is:

(A) Triclinic
(B) Monoclinic
(C) Hexagonal
(D) Rhombohedral

12. In a Simple Cubic (SC) unit cell, the total effective number of atoms per unit cell is:

(A) 1 (8 * 1/8)
(B) 2
(C) 4
(D) 6

13. The Coordination Number (number of equidistant nearest neighbors) for a Simple Cubic lattice is:

(A) 6
(B) 8
(C) 12
(D) 4

14. The Atomic Packing Factor (APF) of a Simple Cubic crystal structure is approximately:

(A) 0.52 (52 %)
(B) 0.68 (68 %)
(C) 0.74 (74 %)
(D) 0.34 (34 %)

15. In a Body-Centered Cubic (BCC) unit cell, the total effective number of atoms per cell is:

(A) 2 (8 * 1/8 + 1)
(B) 1
(C) 4
(D) 8

16. The Coordination Number for a Body-Centered Cubic (BCC) lattice is:

(A) 8
(B) 6
(C) 12
(D) 4

17. The Atomic Packing Factor (APF) of a Body-Centered Cubic (BCC) crystal is:

(A) 0.68 (68 %)
(B) 0.52 (52 %)
(C) 0.74 (74 %)
(D) 0.78 (78 %)

18. In a Face-Centered Cubic (FCC) unit cell, the total effective number of atoms per unit cell is:

(A) 4 (8 * 1/8 + 6 * 1/2)
(B) 2
(C) 1
(D) 6

19. The Coordination Number for a Face-Centered Cubic (FCC) lattice is:

(A) 12
(B) 8
(C) 6
(D) 4

20. The Atomic Packing Factor (APF) of a Face-Centered Cubic (FCC) structure is:

(A) 0.74 (74 %, close-packed)
(B) 0.68 (68 %)
(C) 0.52 (52 %)
(D) 0.82 (82 %)

21. Which of the following common engineering metals crystallizes in a Face-Centered Cubic (FCC) lattice at room temperature?

(A) Copper (Cu) and Aluminum (Al)
(B) alpha-Iron (Ferrite)
(C) Tungsten (W)
(D) Chromium (Cr)

22. Which of the following metals exhibits a Body-Centered Cubic (BCC) crystal structure at room temperature?

(A) alpha-Iron (Ferrite)
(B) Copper (Cu)
(C) Gold (Au)
(D) Silver (Ag)

23. The relation between atomic radius r and lattice parameter 'a' for a Face-Centered Cubic (FCC) crystal is:

(A) r = a * sqrt(2) / 4
(B) r = a * sqrt(3) / 4
(C) r = a / 2
(D) r = a * sqrt(3) / 2

24. The relation between atomic radius r and lattice parameter 'a' along the body diagonal of a BCC crystal is:

(A) r = a * sqrt(3) / 4
(B) r = a * sqrt(2) / 4
(C) r = a / 2
(D) r = a / 4

25. Miller indices (h k l) of a crystal plane are defined as the smallest integers proportional to:

(A) Reciprocals of the plane's fractional intercepts on the coordinate axes
(B) Direct intercepts on coordinate axes
(C) Square roots of plane areas
(D) Logarithms of coordinates

26. If a crystal plane is strictly parallel to a crystallographic axis, its corresponding Miller index along that axis is:

(A) 0 (reciprocal of infinity)
(B) 1
(C) Infinity
(D) -1

27. The interplanar spacing d_hkl for adjacent parallel planes (hkl) in a cubic crystal of lattice parameter 'a' is:

(A) d = a / sqrt(h^2 + k^2 + l^2)
(B) d = a * sqrt(h^2 + k^2 + l^2)
(C) d = a / (h + k + l)
(D) d = sqrt(h^2 + k^2 + l^2) / a

28. Bragg's Law of X-ray diffraction from crystal planes separated by distance 'd' is formulated as:

(A) 2 * d * sin(theta) = n * lambda
(B) d * sin(theta) = n * lambda
(C) 2 * d * cos(theta) = n * lambda
(D) d * tan(theta) = 2 * n * lambda

29. In Bragg's diffraction law (2*d*sin(theta) = n*lambda), the angle 'theta' is known as the:

(A) Glancing (Bragg) angle between incident beam and crystal plane
(B) Angle between incident and reflected beams
(C) Angle with plane normal
(D) Brewster angle

30. The classical Drude-Lorentz model of metals treats valence electrons as an ideal gas obeying:

(A) Maxwell-Boltzmann classical statistics
(B) Fermi-Dirac quantum statistics
(C) Bose-Einstein statistics
(D) Relativistic Dirac statistics

31. The electrical conductivity sigma of a metal in the Drude free electron model is given by:

(A) sigma = n * e^2 * tau / m
(B) sigma = n * e * tau / m
(C) sigma = n * e^2 / (m * tau)
(D) sigma = m / (n * e^2 * tau)

32. In the conductivity expression sigma = n*e^2*tau/m, the parameter 'tau' physically represents:

(A) Mean relaxation time between consecutive electron collisions
(B) Transmission coefficient
(C) Thermal conductivity
(D) Lattice period

33. The Wiedemann-Franz Law asserts that the ratio of thermal conductivity K to electrical conductivity sigma is:

(A) Directly proportional to absolute temperature T: K / (sigma * T) = Lorenz number L
(B) Inversely proportional to T
(C) Independent of temperature
(D) Proportional to T^2

34. The quantum mechanical distribution function governing electron occupancy of energy states is:

(A) Fermi-Dirac distribution function f(E)
(B) Maxwell-Boltzmann distribution
(C) Bose-Einstein distribution
(D) Planck radiation law

35. The Fermi-Dirac distribution function f(E) is mathematically formulated as:

(A) f(E) = 1 / [exp((E - E_F) / (k_B * T)) + 1]
(B) f(E) = 1 / [exp((E - E_F) / (k_B * T)) - 1]
(C) f(E) = exp(- (E - E_F) / (k_B * T))
(D) f(E) = 1 / [1 + (E / E_F)]

36. At absolute zero temperature (T = 0 K), the probability of finding an electron in an energy state E below the Fermi level (E < E_F) is:

(A) 1.0 (100 % occupied)
(B) 0.0
(C) 0.5
(D) Infinite

37. At absolute zero temperature (T = 0 K), the occupation probability for states above Fermi level (E > E_F) is:

(A) 0.0 (completely unoccupied)
(B) 1.0
(C) 0.5
(D) 0.25

38. At any finite non-zero temperature (T > 0 K), the probability of electron occupancy exactly at the Fermi energy level (E = E_F) is:

(A) 0.5 (exactly 50 %)
(B) 1.0
(C) 0.0
(D) Depends on material density

39. Bloch's Theorem states that the wave function of an electron moving in a periodic crystal lattice potential is:

(A) A plane wave modulated by a function possessing the periodicity of the lattice: psi_k(r) = exp(i*k*r) * u_k(r)
(B) A pure spherical standing wave
(C) A decaying exponential wave
(D) A constant uniform amplitude

40. According to the Kronig-Penney model, the origin of forbidden energy band gaps in crystalline solids arises from:

(A) Bragg reflection of electron matter waves by the periodic crystal lattice potential
(B) Coulomb repulsion between valence electrons
(C) Nuclear radioactive decay
(D) Thermal phonon collisions

41. A point defect created when an atom is missing from its normal crystallographic lattice site is called a:

(A) Vacancy
(B) Interstitial
(C) Frenkel defect
(D) Dislocation

42. A defect pair consisting of a cation vacancy and an equal number of anion vacancies preserving electrical neutrality is a:

(A) Schottky defect
(B) Frenkel defect
(C) Edge dislocation
(D) Interstitial defect

43. A defect formed when an ion is displaced from its regular lattice site into an adjacent interstitial position is a:

(A) Frenkel defect
(B) Schottky defect
(C) Grain boundary
(D) Screw dislocation

44. The Burgers vector b of an Edge Dislocation is oriented:

(A) Perpendicular to the dislocation line
(B) Parallel to the dislocation line
(C) At 45 degrees to dislocation line
(D) Inversely to dislocation line

45. The Burgers vector b of a Screw Dislocation is oriented:

(A) Strictly parallel to the dislocation line
(B) Perpendicular to the dislocation line
(C) Along the slip plane normal
(D) Randomly oriented

46. A planar 2D crystalline defect separating two adjacent single-crystal grains of different crystallographic orientations is a:

(A) Grain Boundary
(B) Vacancy
(C) Screw dislocation
(D) Frenkel defect

47. For a cubic crystal with lattice constant a = 0.40 nm, the interplanar spacing for the (1 0 0) planes is:

(A) 0.40 nm (a / 1)
(B) 0.28 nm
(C) 0.23 nm
(D) 0.20 nm

48. For the cubic crystal in Q47 (a = 0.40 nm), the interplanar spacing d for the (1 1 0) planes is:

(A) 0.283 nm (a / sqrt(2))
(B) 0.400 nm
(C) 0.231 nm
(D) 0.200 nm

49. For the cubic crystal in Q47 (a = 0.40 nm), the interplanar spacing d for the (1 1 1) planes is:

(A) 0.231 nm (a / sqrt(3))
(B) 0.400 nm
(C) 0.283 nm
(D) 0.133 nm

50. A monochromatic X-ray beam of wavelength lambda = 0.154 nm undergoes first-order (n=1) Bragg reflection from crystal planes with d = 0.282 nm. The glancing angle theta is:

(A) 15.86 degrees (sin(theta) = 0.154 / (2*0.282))
(B) 30.00 degrees
(C) 45.00 degrees
(D) 8.25 degrees

51. If second-order (n=2) diffraction is observed for the setup in Q50, the glancing angle theta_2 satisfies:

(A) sin(theta_2) = 2 * 0.154 / (2 * 0.282) = 0.5461 => theta_2 = 33.10 degrees
(B) 31.72 degrees
(C) 15.86 degrees
(D) 45.00 degrees

52. In a solid with band gap E_g = 1.1 eV (such as Silicon), the material behaves electronically as a:

(A) Semiconductor
(B) Metal conductor
(C) Wide-gap insulator
(D) Superconductor

53. In Diamond, the energy band gap between valence band and conduction band is approximately E_g = 5.4 eV. Diamond is classified as an:

(A) Insulator
(B) Semiconductor
(C) Good conductor
(D) Semimetal

54. In metallic conductors such as Copper and Silver, the electrical conductivity is extremely high because:

(A) The conduction band is partially filled or overlaps with the valence band (E_g = 0)
(B) The band gap E_g is greater than 10 eV
(C) All electrons are immobilized in deep potential wells
(D) Covalent bonds lock all valence electrons

55. The effective mass m* of an electron in a periodic crystal lattice band is defined mathematically by:

(A) m* = h_bar^2 / (d^2E / dk^2)
(B) m* = h_bar * (dE / dk)
(C) m* = (d^2E / dk^2) / h_bar^2
(D) m* = h_bar^2 * k / E

56. Near the top of an energy band where curvature d^2E/dk^2 is negative, the effective mass m* of an electron is:

(A) Negative (behaving physically as a positive mobile charge carrier or Hole)
(B) Positive
(C) Zero
(D) Infinite

57. At the inflection point of an E-k energy dispersion curve where d^2E/dk^2 = 0, the effective mass m* becomes:

(A) Infinite
(B) Zero
(C) 1.0
(D) Negative

58. In intrinsic semiconductors at room temperature (T = 300 K), the Fermi energy level E_F lies:

(A) Approximately at the exact center of the forbidden energy band gap: E_F = (E_c + E_v) / 2
(B) Inside the conduction band
(C) At the top of the valence band
(D) Far below the valence band

59. When an intrinsic semiconductor is doped with trivalent impurity atoms (such as Boron or Gallium), it forms a:

(A) p-type semiconductor (acceptor levels near valence band)
(B) n-type semiconductor
(C) Metallic conductor
(D) Dielectric insulator

60. When an intrinsic semiconductor is doped with pentavalent impurity atoms (such as Phosphorus or Arsenic), it forms an:

(A) n-type semiconductor (donor levels near conduction band)
(B) p-type semiconductor
(C) Insulator
(D) Amorphous glass

61. The phenomenon wherein certain materials exhibit EXACTLY ZERO electrical resistance below a critical temperature T_c is termed:

(A) Superconductivity
(B) Semiconductivity
(C) Piezoelectricity
(D) Ferroelectricity

62. The Meissner Effect in superconductors refers to:

(A) The complete expulsion of internal magnetic flux (B = 0) upon transitioning below T_c
(B) A sudden jump in electrical resistivity
(C) The spontaneous emission of photons
(D) The emission of acoustic phonons

63. Due to the Meissner effect, an ideal superconductor in the superconducting state acts as a:

(A) Perfect diamagnetic material with magnetic susceptibility chi = - 1
(B) Paramagnetic material with chi = + 1
(C) Ferromagnetic material with large positive chi
(D) Antiferromagnetic substance

64. The critical magnetic field H_c(T) required to destroy superconductivity at temperature T is given by Tuyn's empirical relation:

(A) H_c(T) = H_c(0) * [1 - (T / T_c)^2]
(B) H_c(T) = H_c(0) * [1 + (T / T_c)^2]
(C) H_c(T) = H_c(0) * [1 - T / T_c]
(D) H_c(T) = H_c(0) / (1 - T / T_c)

65. Type I superconductors are characterized by:

(A) A single sharp critical magnetic field H_c and complete Meissner expulsion until abrupt transition to normal state
(B) Two critical fields H_c1 and H_c2 with a mixed vortex state
(C) High critical temperatures exceeding 100 K
(D) High current carrying capacity in strong fields

66. Type II superconductors possess two critical magnetic fields (H_c1 and H_c2). In the intermediate regime between H_c1 and H_c2, the material exists in the:

(A) Vortex (Mixed / Shubnikov) state with partial magnetic flux penetration
(B) Normal resistive state
(C) Completely diamagnetic Meissner state
(D) Ferromagnetic saturation state

67. The microscopic BCS Theory (Bardeen-Cooper-Schrieffer, 1957) explains conventional superconductivity as arising from:

(A) Formation of electron pairs (Cooper Pairs) bound by attractive phonon-mediated electron-lattice interactions
(B) Direct electrostatic attraction between electrons
(C) Magnetic dipole alignment of electron spins
(D) Relativistic contraction of the lattice

68. A Cooper pair consists of two electrons possessing:

(A) Opposite momenta and opposite spins: (k, up-spin) and (-k, down-spin)
(B) Identical momenta and parallel spins
(C) Arbitrary random orientations
(D) Orthogonal velocities and parallel spins

69. Because Cooper pairs have integer net spin (S = 0), they behave collectively as:

(A) Bosons, condensing into a single macroscopic ground quantum state
(B) Fermions obeying Pauli exclusion
(C) Classical non-interacting particles
(D) High-energy plasma ions

70. The critical temperature T_c for Mercury (Hg), discovered by Heike Kamerlingh Onnes in 1911, is:

(A) 4.2 Kelvin
(B) 9.2 Kelvin
(C) 77 Kelvin
(D) 1.2 Kelvin

71. High-temperature superconductors (HTS) such as YBa2Cu3O7-x (YBCO) have critical temperatures T_c around 93 K, allowing cooling via inexpensive:

(A) Liquid Nitrogen (boiling point 77 K)
(B) Liquid Helium (boiling point 4.2 K)
(C) Liquid Hydrogen (boiling point 20 K)
(D) Dry ice (195 K)

72. The London Penetration Depth lambda_L in a superconductor physically measures:

(A) The characteristic exponential decay distance of external magnetic field into the superconductor surface
(B) The thickness of the superconducting film
(C) The size of a Cooper pair
(D) The lattice parameter of the crystal

73. The BCS Coherence Length xi measures:

(A) The characteristic spatial separation / size of the two electrons in a Cooper pair (~10^-7 to 10^-6 m)
(B) The mean free path of normal electrons
(C) The lattice vibration amplitude
(D) The unit cell edge length

74. A Josephson Junction consists of two superconducting electrodes separated by an ultra-thin:

(A) Insulating barrier (~ 1 to 2 nm thick) through which Cooper pairs tunnel quantum mechanically
(B) Thick copper conductor plate
(C) Semiconducting p-n junction
(D) Ferromagnetic spacer

75. SQUID (Superconducting Quantum Interference Device) sensors exploit the Josephson effect and flux quantization to measure:

(A) Extremely minuscule magnetic fields as low as 10^-15 Tesla (Femtotesla)
(B) High voltage power surges
(C) Ultrasound wave frequencies
(D) Laser coherence length

76. Copper has an FCC crystal structure with lattice parameter a = 0.361 nm (atomic mass M = 63.54 g/mol, N_A = 6.022 * 10^23 /mol). The theoretical crystal density is:

(A) 8930 kg/m^3 (8.93 g/cm^3)
(B) 7870 kg/m^3
(C) 2700 kg/m^3
(D) 19300 kg/m^3

77. For alpha-iron (BCC, a = 0.287 nm), the interplanar spacing for the most closely packed slip planes (1 1 0) is:

(A) 0.203 nm (0.287 / sqrt(2))
(B) 0.287 nm
(C) 0.166 nm
(D) 0.101 nm

78. The Miller indices of a plane that makes intercepts of 2a, 3b, and is parallel to the c-axis are:

(A) (3 2 0)
(B) (2 3 0)
(C) (3 2 1)
(D) (2 3 infinity)

79. The Miller indices of a crystallographic direction vector connecting origin (0, 0, 0) to point (1/2, 1, 0) are:

(A) [1 2 0]
(B) [2 1 0]
(C) (1 2 0)
(D) <1 2 0>

80. In a cubic crystal, the angle theta between crystallographic directions [1 0 0] and [1 1 1] is calculated as:

(A) 54.74 degrees (cos(theta) = 1 / sqrt(3))
(B) 45.00 degrees
(C) 90.00 degrees
(D) 35.26 degrees

81. In X-ray diffraction of an FCC lattice, diffraction peaks occur ONLY for Miller indices (h k l) where:

(A) h, k, l are all odd or all even (unmixed indices)
(B) h + k + l = even integer
(C) h, k, l are all positive
(D) h^2 + k^2 + l^2 <= 10

82. In X-ray diffraction of a BCC lattice, diffraction reflection peaks are strictly observed ONLY when:

(A) h + k + l is an even integer
(B) h, k, l are all odd
(C) h, k, l are all even
(D) h * k * l is non-zero

83. The Fermi energy of Silver at T = 0 K is E_F = 5.48 eV. The corresponding Fermi velocity v_F of conduction electrons is (m_e = 9.109 * 10^-31 kg):

(A) 1.39 * 10^6 m/s (sqrt(2 * E_F / m_e))
(B) 3.00 * 10^8 m/s
(C) 2.18 * 10^5 m/s
(D) 9.81 * 10^4 m/s

84. For Silver in Q83 (E_F = 5.48 eV), the Fermi Temperature T_F = E_F / k_B is approximately (k_B = 8.617 * 10^-5 eV/K):

(A) 63,600 Kelvin
(B) 300 Kelvin
(C) 12,500 Kelvin
(D) 100,000 Kelvin

85. Because the Fermi temperature T_F (~60,000 K) is vastly higher than room temperature (~300 K):

(A) Only a tiny fraction (~ T / T_F ~ 1%) of conduction electrons near E_F participate in thermal absorption and electrical conduction
(B) All conduction electrons absorb heat equally obeying classical 3/2 k_B
(C) The electrons freeze into fixed positions
(D) Ohm's law breaks down completely

86. The electronic contribution to the molar specific heat C_v_el of metals at low temperatures varies with temperature as:

(A) Linear with temperature: C_v_el = gamma_Sommerfeld * T
(B) Proportional to T^3 (Debye law)
(C) Constant at 3 * R
(D) Proportional to exp(- Delta / k_B*T)

87. The lattice vibration (phonon) contribution to the specific heat C_v_ph of crystalline solids at low temperatures obeys:

(A) Debye T^3 law: C_v_ph = A * T^3
(B) Sommerfeld linear law C_v = gamma * T
(C) Einstein exponential law
(D) Dulong-Petit constant 3*R

88. In the 1D Kronig-Penney model, when the barrier strength parameter P -> infinity, the energy band spectrum degenerates into:

(A) Discrete atomic-like energy levels identical to a particle in an infinite potential box
(B) A completely continuous energy spectrum
(C) Zero allowed states
(D) Overlapping semi-metallic bands

89. When the barrier parameter P -> 0 in the Kronig-Penney model, the electron energy spectrum corresponds to:

(A) Completely free electrons with parabolic dispersion: E(k) = h_bar^2 * k^2 / (2 * m)
(B) Discrete isolated delta levels
(C) Wide band gaps everywhere
(D) Infinite effective mass

90. The first Brillouin Zone boundary for a 1D lattice of spacing 'a' is located in k-space at:

(A) k = + pi / a and k = - pi / a
(B) k = + 2*pi / a and k = - 2*pi / a
(C) k = 0
(D) k = + pi / (2*a)

91. At the first Brillouin Zone boundary (k = pi / a), the electron group velocity v_g = (1 / h_bar) * (dE / dk) is:

(A) Zero (forming standing waves due to Bragg reflection)
(B) Speed of light c
(C) Fermi velocity v_F
(D) Infinite

92. An electron moves in a 1D conduction band with energy dispersion E(k) = E0 - 2 * gamma * cos(k * a). The effective mass m* at band bottom (k = 0) is:

(A) m* = h_bar^2 / (2 * gamma * a^2)
(B) m* = - h_bar^2 / (2 * gamma * a^2)
(C) m* = h_bar^2 / (gamma * a)
(D) m* = 0

93. For the energy dispersion in Q92, the effective mass m* at the top of the band (k = pi / a) is:

(A) m* = - h_bar^2 / (2 * gamma * a^2) (Negative effective mass)
(B) + h_bar^2 / (2 * gamma * a^2)
(C) Infinite
(D) Zero

94. A semiconductor sample of thickness t = 0.5 mm carrying current I = 10 mA in magnetic field B = 0.5 T perpendicular to current exhibits Hall voltage V_H = 2.0 mV. The Hall coefficient R_H is:

(A) R_H = (V_H * t) / (I * B) = (2.0*10^-3 * 0.5*10^-3) / (10*10^-3 * 0.5) = 2.0 * 10^-4 m^3/C
(B) 1.0 * 10^-4 m^3/C
(C) 4.0 * 10^-4 m^3/C
(D) 0.5 * 10^-4 m^3/C

95. For the semiconductor in Q94, the positive sign of the measured Hall coefficient R_H indicates that the predominant charge carriers are:

(A) Holes (p-type semiconductor)
(B) Electrons (n-type semiconductor)
(C) Ions
(D) Neutrons

96. A Lead (Pb) superconductor has critical field H_c(0) = 6.4 * 10^4 A/m at 0 K and critical temperature T_c = 7.2 K. The critical field at T = 4.2 K is:

(A) 4.22 * 10^4 A/m (6.4*10^4 * [1 - (4.2/7.2)^2])
(B) 6.40 * 10^4 A/m
(C) 2.18 * 10^4 A/m
(D) 5.15 * 10^4 A/m

97. The energy gap Delta(0) in the electronic density of states of a BCS superconductor at T = 0 K is related to its critical temperature T_c by:

(A) 2 * Delta(0) = 3.52 * k_B * T_c
(B) 2 * Delta(0) = 1.0 * k_B * T_c
(C) Delta(0) = 3.52 * k_B * T_c
(D) 2 * Delta(0) = k_B * T_c / 3.52

98. Magnetic flux trapped within a hollow superconducting cylinder is quantized in discrete integer multiples of the flux quantum Phi_0:

(A) Phi_0 = h / (2 * e) ~= 2.0678 * 10^-15 Weber (Tesla-m^2)
(B) Phi_0 = h / e
(C) Phi_0 = h_bar / (2 * e)
(D) Phi_0 = 2 * h / e

99. In the DC Josephson Effect, a constant DC supercurrent I_s flows across an insulating junction with zero applied voltage, given by:

(A) I_s = I_c * sin(Delta phi), where Delta phi is the quantum phase difference
(B) I_s = I_c * cos(Delta phi)
(C) I_s = I_c * Delta phi
(D) I_s = V / R

100. In the AC Josephson Effect, applying a constant DC voltage V across the junction induces an ultra-precise high-frequency oscillating AC supercurrent with frequency:

(A) f_J = (2 * e * V) / h = V / Phi_0 ~= 483.6 GHz per millivolt
(B) f_J = (e * V) / h
(C) f_J = h / (2 * e * V)
(D) f_J = (2 * e) / (h * V)

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Define a Crystal Lattice, Basis, and Unit Cell. State the seven crystal systems in three dimensions along with their axial length and interaxial angle relationships.

Q2. Derive the Atomic Packing Factor (APF), Coordination Number, effective number of atoms per unit cell, and atomic radius relation with lattice constant for Simple Cubic (SC), Body-Centered Cubic (BCC), and Face-Centered Cubic (FCC) crystal structures.

Q3. Derive the general expression for the theoretical density rho of a crystalline solid in terms of its lattice parameter 'a', effective number of atoms per unit cell N, atomic mass M, and Avogadro's number N_A. Calculate the density of Face-Centered Cubic Copper (M = 63.54 g/mol, a =

Q4. Explain the procedure for determining Miller Indices (h k l) of a crystal plane. Derive the formula for interplanar spacing d_hkl for a cubic crystal lattice.

Q5. State Bragg's Law of X-ray diffraction. Derive the mathematical condition for constructive interference of X-rays scattered from parallel crystal planes.

Q6. Monochromatic X-rays of wavelength lambda = 0.154 nm (Cu-K_alpha radiation) are incident on a cubic crystal of lattice parameter a = 0.408 nm (Gold). Calculate: (1) Interplanar spacings for (1 1 1), (2 0 0), and (2 2 0) planes, (2) First-order (n = 1) Bragg diffraction glancing angles theta

Q7. Classify Crystal Imperfections (Lattice Defects) based on their dimensionality: Point defects, Line defects, Planar defects, and Volume defects. Explain Schottky and Frenkel defects with neat schematic representations.

Q8. Outline the Classical Free Electron Theory (Drude-Lorentz Model). State its basic postulates, derive the expression for electrical conductivity sigma = n*e^2*tau/m, and explain its major experimental failures.

Q9. State the postulates of Sommerfeld's Quantum Free Electron Theory. Explain the Fermi-Dirac Distribution Function and plot its behavior at T = 0 K and T > 0 K. Define Fermi Energy, Fermi Velocity, and Fermi Temperature.

Q10. Explain the Band Theory of Solids based on the Kronig-Penney Model. Discuss how the periodic potential of a crystal splits atomic energy levels into allowed energy bands and forbidden band gaps.

Q11. Distinguish between Conductors, Semiconductors, and Insulators on the basis of Band Theory. Draw representative energy band diagrams illustrating the conduction band, valence band, and forbidden band gap for each category.

Q12. Derive the expression for the Effective Mass m* of an electron moving in a crystal energy band. Explain the physical significance of positive, negative, and infinite effective mass.

Q13. Explain the Hall Effect in semiconductors. Derive the expressions for Hall Voltage V_H, Hall Coefficient R_H, and carrier concentration 'n'. Explain how Hall effect measurements distinguish between n-type and p-type semiconductors.

Q14. Define Superconductivity. Describe the Meissner Effect and explain why it proves that a superconductor is a perfect diamagnet and not merely an ideal conductor with zero resistance.

Q15. Compare Type I and Type II Superconductors with respect to: Critical magnetic fields, Meissner effect behavior, Mixed (vortex) state, and Practical engineering applications.

Q16. Explain the BCS Theory of Superconductivity. Describe the formation of Cooper Pairs via electron-phonon-electron interactions and explain how energy gap Delta(T) accounts for zero resistivity.

Q17. Describe the DC and AC Josephson Effects across a Superconductor-Insulator-Superconductor (S-I-S) tunnel junction. State the fundamental equations for supercurrent and Josephson frequency.

Q18. A superconducting solenoid magnet constructed of Nb-Ti wire (T_c = 9.2 K, H_c(0) = 1.2 * 10^7 A/m) operates at liquid helium temperature T = 4.2 K. Calculate: (1) The critical magnetic field H_c at 4.2 K, (2) If the solenoid has 2500 turns per meter, compute the maximum allowable operating

Q19. Explain the concept of Magnetic Flux Quantization in a superconducting ring. Derive the expression for the Magnetic Flux Quantum Phi_0 = h / (2*e) and state its numerical value.

Q20. Explain High-Temperature Superconductors (HTS). Discuss their discovery, crystalline structure (cuprate perovskites), critical temperatures, and their practical engineering significance compared to conventional Low-Tc superconductors.

Q21. Calculate the Fermi energy E_F, Fermi velocity v_F, and Fermi temperature T_F for Copper, assuming each Cu atom contributes one free conduction electron. Given: atomic mass M = 63.54 g/mol, density rho = 8.94 g/cm^3, m_e = 9.109 * 10^-31 kg, h = 6.626 * 10^-34 J-s.

Q22. Explain the Brillouin Zones in reciprocal lattice space. Derive the first Brillouin zone boundaries for a 1D linear lattice of constant 'a', and sketch the E-k dispersion curves showing energy discontinuities.

Q23. A semiconductor crystal has an energy band dispersion described by the tight-binding relation: E(k) = E0 - 2 * t * cos(k * a), where t = 1.2 eV and a = 0.30 nm. Calculate: (1) Width of the allowed conduction band, (2) Electron group velocity at k = pi / (2*a), (3) Effective mass m* at

Q24. State and prove Bloch's Theorem for an electron moving in a one-dimensional periodic crystal potential V(x) = V(x + a). Explain the properties of Bloch wave functions.

Q25. Explain the London Equations of Superconductivity. Derive the London Penetration Depth lambda_L and show how it explains the exponential decay of magnetic flux inside a superconductor.

Q26. A superconducting material has a London penetration depth of lambda_L(0) = 40 nm at T = 0 K and a critical temperature T_c = 9.0 K. Using Gorter-Casimir two-fluid model relation lambda_L(T) = lambda_L(0) / sqrt(1 - (T/T_c)^4), compute the penetration depth at T = 4.5 K and T = 8.5 K.

Q27. Explain the Powder Method (Debye-Scherrer Technique) of X-ray crystal structure analysis. Describe the experimental setup, formation of diffraction cones, and how lattice parameters are indexed.

Q28. Explain the origin of Energy Bands in periodic solids using the LCAO (Linear Combination of Atomic Orbitals) tight-binding approach. Contrast the electronic structures of sodium (metal) and carbon (diamond insulator).

Q29. A SQUID magnetometer is used to measure magnetic field variations in structural health monitoring of reinforced concrete bridges. The SQUID sensing loop has an area A = 2.0 mm^2. Calculate: (1) The magnetic field change Delta B corresponding to the passage of exactly one magnetic flux

Q30. Summarize the comprehensive engineering applications of Solid State Physics and Superconductivity in modern Civil and Infrastructure Engineering: (1) High-strength structural alloys and rebar crystallography, (2) Semiconductor sensors and smart structural monitoring, (3) Superconducting

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- Atomic Packing Factor (APF):
APF = [2 * (4/3)*pi*r^3] / a^3 = [(8/3)*pi * (a*sqrt(3)/4)^3] / a^3 = [(8/3)*pi *
(3*sqrt(3)/64)] = pi * sqrt(3) / 8 = 0.6802 =~ 68.0%.

3. Face-Centered Cubic (FCC) Structure:
- Atoms per cell: 8 corners * (1/8) + 6 face centers * (1/2) = 1 + 3 = 4 atoms/cell.
- Coordination Number (CN): In the face plane 4 neighbors, 4 in plane above, 4 in plane below
=> CN = 12.
- Nearest neighbor contact: Atoms touch along face diagonal of length a*sqrt(2):
4 * r = a * sqrt(2) => r = a * sqrt(2) / 4.
- Atomic Packing Factor (APF):
APF = [4 * (4/3)*pi*r^3] / a^3 = [(16/3)*pi * (a*sqrt(2)/4)^3] / a^3 = [(16/3)*pi *
(2*sqrt(2)/64)] = pi * sqrt(2) / 6 = 0.7405 =~ 74.0%.
Conclusion: FCC is close-packed (74.0%), BCC is moderately packed (68.0%), and SC is loosely
packed (52.4%).

- Step 3 (Take Reciprocals): Invert each intercept: 1/p, 1/q, 1/r. (1/infinity = 0).
- Step 4 (Clear Fractions): Multiply the reciprocals by their Least Common Denominator (LCD) to yield the smallest set of coprime integers: h, k, l.
- Step 5 (Format): Enclose the integers in parentheses without commas: (h k l). If an intercept is negative, place a bar over the index: (h bar-k l).

2. Derivation of Interplanar Spacing d_hkl in Cubic Lattices:
Consider plane ABC belonging to family (hkl) closest to origin O.
Intercepts on axes: OA = a/h, OB = a/k, OC = a/l.
Let ON be the perpendicular normal drawn from origin O to plane ABC. The length ON = d_hkl.
Let direction cosines of normal ON with axes x, y, z be cos(alpha), cos(beta), cos(gamma):
cos(alpha) = d / OA = d / (a/h) = (d * h) / a.
cos(beta) = d / OB = d / (a/k) = (d * k) / a.
cos(gamma) = d / OC = d / (a/l) = (d * l) / a.
By the fundamental trigonometric identity of direction cosines in orthogonal 3D space:
cos^2(alpha) + cos^2(beta) + cos^2(gamma) = 1.
[(d * h) / a]^2 + [(d * k) / a]^2 + [(d * l) / a]^2 = 1.
(d^2 / a^2) * [h^2 + k^2 + l^2] = 1.
d^2 = a^2 / (h^2 + k^2 + l^2) => d_hkl = a / sqrt(h^2 + k^2 + l^2).

>> **Solution to Question 5:** Bragg's Law of X-ray diffraction, Derive the mathematical condition for constructive interference of X-rays scattered from parallel

* Density Effect: Density of crystal DECREASES noticeably because volume remains constant while mass is lost.
* Favored In: Highly ionic crystals with high coordination numbers and similar cation/anion radii (e.g., NaCl, KCl, CsCl).
- Frenkel Defect:
* Mechanism: An ion (usually the smaller cation) leaves its normal lattice site and lodges in an interstitial void, creating a vacancy-interstitial pair.
* Electrical Neutrality: Perfectly preserved.
* Density Effect: Overall crystal density remains COMPLETELY UNCHANGED (mass and volume invariant).
* Favored In: Crystals with large differences in ionic radii and low coordination numbers (e.g., AgCl, ZnS).

- Conduction electrons are treated as quantum mechanical waves confined in a 3D potential well representing the metal boundaries.
- Electrons obey the Pauli Exclusion Principle: each discrete quantum state can hold at most 2 electrons with opposite spins (+1/2, -1/2).
- Electron energy states are occupied according to Fermi-Dirac Quantum Statistics, replacing classical Maxwell-Boltzmann statistics.

2. Fermi-Dirac Distribution Function f(E):
The probability f(E) that a quantum state of energy E is occupied by an electron at temperature T is:
f(E) = 1 / [exp((E - E_F) / (k_B * T)) + 1].

3. Temperature Profiles:
- At T = 0 Kelvin (Absolute Zero):
* For E < E_F: (E - E_F) < 0 => exp(- infinity) = 0 => f(E) = 1 / (0 + 1) = 1.0 (Completely filled).
* For E > E_F: (E - E_F) > 0 => exp(+ infinity) = infinity => f(E) = 1 / infinity = 0.0 (Completely empty).
* The function is a perfect sharp rectangular step at E = E_F.
- At T > 0 Kelvin (Finite Temperature):
* Thermal excitation kicks a small band of electrons (~ k_B*T wide) from below E_F to above E_F.
* Exactly at E = E_F: f(E_F) = 1 / [exp(0) + 1] = 1 / (1 + 1) = 0.5 (Exactly 50% probability at all temperatures).

4. Key Fermi Parameters:
- Fermi Energy E_F: The highest kinetic energy state filled by conduction electrons at absolute zero temperature (T = 0 K).
Formula in 3D: E_F0 = (h_bar^2 / (2*m)) * (3 * pi^2 * n)^(2/3), where n = N/V is electron density.
- Fermi Velocity v_F: Speed of electrons residing at the Fermi surface: v_F = sqrt(2 * E_F / m).
- Fermi Temperature T_F: Temperature at which thermal energy matches Fermi energy: T_F = E_F / k_B.

>> Solution to Question 10: Explain the Band Theory of Solids based on the Kronig-Penney Model. Discuss how the periodic potential of a crystal splits atomic ene

Lattice period: d = a + b.
Applying Bloch's theorem psi(x) = e^(ikx) * u(x) to Schrodinger's equation and taking the delta-function barrier limit
(b -> 0, V0 -> infinity such that P = m*V0*b*a / h_bar^2 remains finite):
The transcendental dispersion relation is obtained:
P * [sin(alpha * a) / (alpha * a)] + cos(alpha * a) = cos(k * a),
where alpha = sqrt(2 * m * E) / h_bar, and k is the crystal wave vector.

3. Analysis of Kronig-Penney Dispersion:
- The right side cos(ka) can strictly take values between -1 and +1.
- Whenever the left side function falls within [-1, +1], real solutions for wave vector k exist => Allowed Energy Bands.
- Whenever |Left Side| > 1, no real k exists; electron waves undergo destructive Bragg reflection => Forbidden Energy Band Gaps.
- Band width increases and band gap decreases with increasing energy E (alpha*a).

From quantum relation E = h_bar * omega => omega = E / h_bar:
v_g = (1 / h_bar) * (dE / dk). (Equation 1)
Now apply an external electric field E. The external work done on the electron in time dt produces energy change dE:
dE = F * dx = F * v_g * dt => dE / dt = F * v_g.
Also, by chain rule: dE / dt = (dE / dk) * (dk / dt) = (h_bar * v_g) * (dk / dt).
Equating both expressions: F * v_g = h_bar * v_g * (dk / dt) => h_bar * (dk / dt) = F. (Equation 2)
Now differentiate group velocity (Equation 1) with respect to time to obtain acceleration a:
a = dv_g / dt = (1 / h_bar) * d/dt(dE / dk) = (1 / h_bar) * (d^2E / dk^2) * (dk / dt).
Substitute dk/dt = F / h_bar from Equation 2:
a = (1 / h_bar) * (d^2E / dk^2) * (F / h_bar) = [(1 / h_bar^2) * (d^2E / dk^2)] * F.
Comparing with Newton's Second Law a = F / m*:
1 / m* = (1 / h_bar^2) * (d^2E / dk^2) => m* = h_bar^2 / (d^2E / dk^2).

2. Physical Interpretation:
- Bottom of Band (d^2E/dk^2 > 0): Curvature is concave upward => m* > 0. Electron accelerates in direction of applied electric force.
- Top of Band (d^2E/dk^2 < 0): Curvature is concave downward => m* < 0. Electron accelerates OPPOSITE to applied force due to intense Bragg reflection from the periodic lattice. A state with negative effective mass electron is physically equivalent to a positive charge carrier (Hole).
- Inflection Point (d^2E/dk^2 = 0): m* approaches infinity. Electron cannot be accelerated by external fields.

>>Solution to Question 13

Explain the Hall Effect in semiconductors. Derive the expressions for Hall Voltage V_H, Hall Coefficient R_H, and carrier concentration

(Hall coefficient is POSITIVE).

- Measurement of the sign of V_H directly and unambiguously determines whether conduction is governed by electrons or holes.

- Examples: Pure elemental metals like Mercury (Hg), Lead (Pb), Tin (Sn), Aluminum (Al).
- Practical Limitations: Cannot carry large electrical currents or operate in strong magnetic fields without quenching.

2. Type II (Hard) Superconductors:
- Critical Fields: Possesses two critical magnetic fields: Lower critical field H_{c1} and Upper critical field H_{c2}.
- Regimes of Behavior:
* $H < H_{c1}$: Perfect Meissner state ($B = 0$), complete flux expulsion.
* $H_{c1} < H < H_{c2}$: Mixed State (Vortex / Shubnikov State). Magnetic flux penetrates the bulk in the form of quantized flux tubes (vortices) surrounded by circulating supercurrents, while the bulk remains superconducting (zero electrical resistance!).
* $H > H_{c2}$: Normal resistive state.
- Coherence Length vs Penetration Depth: $\lambda_L > \xi$ ($\kappa > 0.707$). Surface energy of boundary is negative, favoring flux line vortex arrays.
- Upper Critical Field: Immensely high H_{c2} (exceeding 20 to 100 Tesla!).
- Examples: Niobium-Titanium (NbTi), Niobium-Tin (Nb_3Sn), High-Tc cuprates like YBCO ($YBa_2Cu_3O_7$).
- Applications: High-field superconducting electromagnets for MRI medical scanners, particle accelerators (CERN LHC), Maglev trains, and fusion reactors (ITER).

>>Solution to Question 16 Explain the BCS Theory of Superconductivity. Describe the formation of Cooper Pairs via electron-phonon-electron interactions and ex

- At T < T_c, thermal phonon energies (~ k_B*T) are insufficient to break Cooper pairs. Hence, Cooper pairs travel through the crystal lattice completely unhindered without scattering: ZERO ELECTRICAL RESISTANCE.

To avoid quenching into the normal resistive state, H must not exceed H_c:
n_turns * I_max = H_c(4.2 K) => I_max = H_c / n_turns.
I_max = (9.499 * 10^6 A/m) / (2500 turns/m) = 3,799.6 Amperes =~ 3.80 kA.
Conclusion: Critical field H_c(4.2 K) = 9.50 * 10^6 A/m (B_c = 11.94 T); Maximum operating current I_max = 3,800 A.

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oxides at 35 K (Nobel Prize 1987).
Shortly thereafter, M.K. Wu and C.W. Chu synthesized YBa2Cu3O7-x (YBCO) with T_c = 93 K, breaking the historic 'liquid nitrogen barrier' (77 K).
Subsequent Bi-Sr-Ca-Cu-O (BSCCO, T_c = 110 K) and Tl-Ba-Ca-Cu-O (T_c = 125 K), and Hg-Ba-Ca-Cu-O (T_c = 134 K, 164 K under pressure) were synthesized.

2. Crystal Architecture (Perovskite Cuprates):
- All high-Tc materials possess layered, highly anisotropic perovskite-derived crystal structures featuring parallel 2D Copper-Oxygen (CuO2) planes.
- Superconducting Cooper pairs are formed and conduct predominantly within these CuO2 sheets (d-wave pairing symmetry).
- Intermediate layers act as charge reservoirs that donate holes or electrons to the CuO2 conduction planes upon chemical doping.

3. Engineering Significance:
- Liquid Nitrogen Cooling: Operating above 77 K eliminates costly, scarce liquid helium cryogenics (Liquid N2 is cheaper than bottled milk, abundant, and has 60x higher latent heat of vaporization).
- Massive Upper Critical Fields: H_c2 exceeds 100 Tesla at low temperatures, enabling ultra-compact high-field magnets for compact fusion reactors (SPARC Tokamak) and high-power wind turbine generators.
- Challenges: As brittle ceramics, fabricating ductile flexible cables requires complex powder-in-tube (PIT) or 2G coated conductor epitaxial tape deposition.

>> Solution to Question 21: Calculate the Fermi energy E_F, Fermi velocity v_F, and Fermi temperature T_F for Copper, assuming each Cu atom contributes one f

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At k = pi / (2*a): ka = pi/2 => sin(pi/2) = 1.0.
dE / dk = 2 * t * a = 2 * (1.2 * 1.602 * 10^-19 J) * (0.30 * 10^-9 m) = 1.1534 * 10^-28 J*m.
v_g = (1.1534 * 10^-28) / (1.05457 * 10^-34 J-s) = 1.0937 * 10^6 m/s =~ 1.09 * 10^6 m/s.

3. Effective Mass m* at Band Bottom (k = 0):
m* = h_bar^2 / (d^2E / dk^2).
d^2E / dk^2 = d/dk [2*t*a*sin(ka)] = 2 * t * a^2 * cos(ka).
At k = 0: cos(0) = 1 => d^2E / dk^2 = 2 * t * a^2.
2 * t * a^2 = 2 * (1.2 * 1.602 * 10^-19 J) * (0.30 * 10^-9 m)^2 = 3.8448 * 10^-19 * 9.0 * 10^-20 = 3.4603 * 10^-38 J*m^2.
m* = (1.05457 * 10^-34)^2 / (3.4603 * 10^-38) = 1.1121 * 10^-68 / 3.4603 * 10^-38 = 3.214 * 10^-31 kg.
In terms of free electron rest mass m_e (9.109 * 10^-31 kg):
m* / m_e = 3.214 * 10^-31 / 9.109 * 10^-31 = 0.353 m_e.
Conclusion: Bandwidth = 4.8 eV; Group velocity = 1.09 * 10^6 m/s; Effective mass = 0.35 m_e.

The London brothers phenomenologically described electrodynamics of superconductors by dividing electrons into normal electrons and superconducting electrons (density n_s).
- Acceleration without resistance: Superconducting electrons move without friction under electric field E:
m * (dv_s / dt) = - e * E.
Current density J_s = - n_s * e * v_s => dJ_s / dt = - n_s * e * (dv_s / dt) = - n_s * e * (- e * E / m).
First London Equation: dJ_s / dt = (n_s * e^2 / m) * E. (Explains zero electrical resistance).

2. Second London Equation (Meissner Effect):
Taking the curl of both sides of the First London Equation:
Curl(dJ_s / dt) = (n_s * e^2 / m) * Curl(E).
Using Faraday's Law Curl(E) = - dB/dt:
d/dt [Curl(J_s) + (n_s * e^2 / m) * B] = 0.
To enforce the Meissner effect (B = 0 in static state), London postulated that the bracketed quantity is not merely constant, but IDENTICALLY ZERO:
Second London Equation: Curl(J_s) = - (n_s * e^2 / m) * B.

3. Derivation of London Penetration Depth lambda_L:
By Maxwell's Ampere law (neglecting displacement current for steady fields): Curl(B) = mu_0 * J_s.
Taking the curl of both sides: Curl(Curl(B)) = mu_0 * Curl(J_s).
Using vector identity Curl(Curl(B)) = Grad(Div B) - Del^2(B), and Gauss's law Div(B) = 0:
- Del^2(B) = mu_0 * [- (n_s * e^2 / m) * B].
Del^2(B) = (mu_0 * n_s * e^2 / m) * B.
Defining London Penetration Depth: lambda_L = sqrt(m / (mu_0 * n_s * e^2)):
Del^2(B) = B / (lambda_L^2).

4. Exponential Field Decay at Surface:
For a semi-infinite superconductor occupying x > 0 with external field B0 applied at surface x = 0:
d^2B / dx^2 = B / lambda_L^2 => B(x) = B0 * exp(- x / lambda_L).
Physical significance: Magnetic field does not vanish discontinuously at the surface, but decays exponentially over skin depth lambda_L (~ 50 nm).

>> Solution to Question 26: A superconducting material has a London penetration depth of lambda_L(0) = 40 nm at T = 0 K and a critical temperature T_c = 9.0 K.

(T / T_c)^4 = (0.94444)^4 = 0.7956.
1 - (T / T_c)^4 = 1 - 0.7956 = 0.2044.
sqrt(0.2044) = 0.4521.
lambda_L(8.5 K) = 40.0 nm / 0.4521 = 88.48 nm.

4. Physical Conclusion:
Near absolute zero (up to T_c/2), penetration depth increases very slightly (only to 41.3 nm).
As temperature approaches T_c (8.5 K), lambda_L more than doubles to 88.5 nm and diverges to infinity at T = T_c (where superconducting carrier density n_s -> 0).

- The crystal wave function is constructed as a linear combination: Psi_crystal = Sigma_i c_i * psi_atomic(r - R_i).
- S-orbitals combine symmetrically (bonding state, lower energy) and antisymmetrically (antibonding state, higher energy).
- For N atoms, N discrete orthogonal molecular orbitals form an Allowed Energy Band of bandwidth Delta E ~ 2*z*t (where z is coordination number and t is transfer hopping integral).

2. Case Study: Sodium (Na, Metal, Z = 11):
- Electron configuration: 1s^2 2s^2 2p^6 3s^1.
- The completely filled 1s, 2s, 2p core levels remain tightly bound with negligible overlap (narrow core bands).
- The outermost 3s orbital has 1 valence electron per atom, but can hold 2 electrons per atom (spin up/down).
- Thus, for N sodium atoms, the 3s band contains 2*N available quantum states but only N electrons.
- The 3s conduction band is EXACTLY HALF-FILLED. Plentiful empty states exist immediately above the Fermi energy.
- Result: Extremely high electrical and thermal conductivity (Typical Good Metal).

3. Case Study: Carbon in Diamond (Insulator, Z = 6):
- Isolated configuration: 1s^2 2s^2 2p^2 (4 valence electrons: two 2s, two 2p).
- In diamond tetrahedral bonding, 2s and three 2p orbitals hybridize into four equivalent sp^3 hybrid orbitals per atom.
- Upon crystal formation, sp^3 orbitals split into: 4*N bonding states (Valence Band) and 4*N antibonding states (Conduction Band).
- All 4*N valence electrons completely fill the lower bonding valence band.
- The conduction band is completely empty, separated by a gigantic band gap E_g = 5.4 eV.
- Result: Exceptional electrical insulator and transparent optical dielectric.

Solution Statement A 29. >>FPSolution Statement A 29: A SQUID magnetometer is used to measure magnetic field variations in structural health monitoring of reinforced concrete bridges. The

Solution Statement Question 30: Summarize the comprehensive engineering applications of Solid State Physics and Superconductivity in modern Civil and Infrastructure